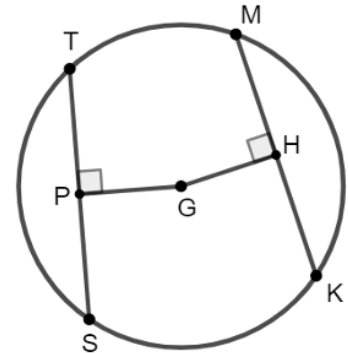


Geometry
Unit 12
Lesson 1 Practice

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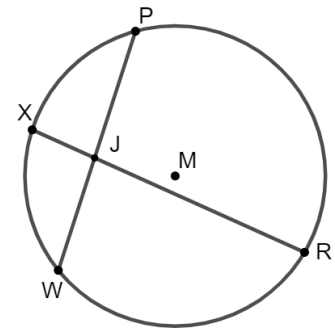
Circle G with points T, M, K , and S on the circle are shown with chords TS and MK equidistant from the center. $TS = (9x + 6)$, $MK = (11x - 10)$, and $m\angle TPG = m\angle MHG = 90^\circ$

- Find the value of x .
- What is the length of chord MK ?



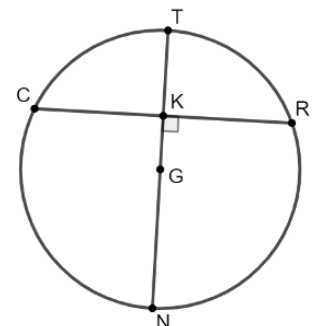
In Circle M , chords RX and PW intersect at point J . Points R, W, X , and P lie on Circle M . $RJ = (8x + 3)$, $JX = 4$, $PJ = (4x - 2)$, and $JW = 12$.

- Solve for the value of x .
- What is the length of line segment RJ ?
- Find the length of chord PW .



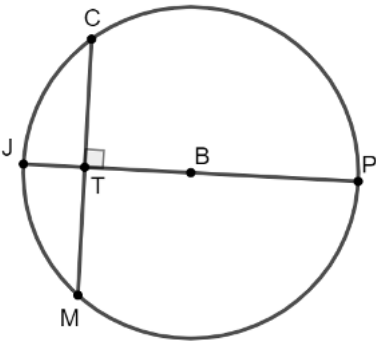
Circle G is shown where points C, T, R , and N are on the circle. The diameter, NT , is a perpendicular bisector of chord CR . $CK = (5x + 6)$ and $KR = (7x - 2)$.

- Calculate the value of x .
- What is the length of chord CR ?



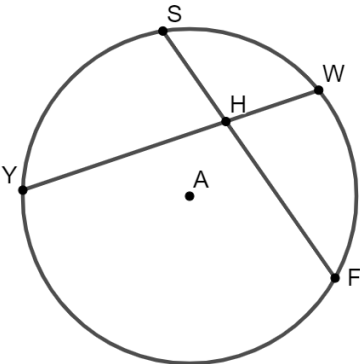
Chord CM is drawn in Circle B so that it intersects the diameter JP .

8. Given that $CM = 24$ and $TB = 5$, what is the length of BP ?



Circle A is shown where points Y, S, W , and F lie on the circle. Chords SF and YW intersect at point H . $YH = (4x + 2)$, $SH = 9$, $HW = 8$, and $HF = (4x)$. Mia and Nazir are solving for x and their work is shown below.

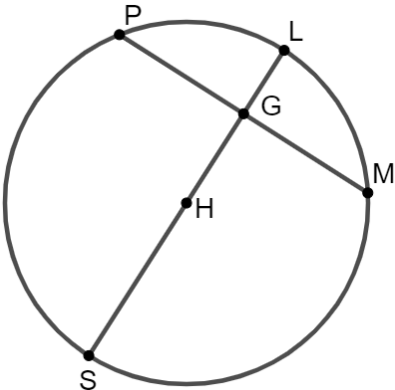
Mia's Work	Nazir's Work
$8(4x + 2) = 9(4x)$ $32x + 2 = 36x$ $4x = 2$ $x = 0.5$	$8(4x + 2) = 9(4x)$ $32x + 16 = 36x$ $4x = 16$ $x = 4$



9. Whose work do you agree with? Justify your answer.

In Circle H , points P, L, M , and S lie on the circle. Chord PM intersects the diameter SL at point G and $m\angle HGP = 90^\circ$.

10. If $PM = 32$ and $HS = 18$, what is the length of GH ?
Round to the nearest hundredth if necessary.



Geometry
Unit 12
Lesson 2 Practice

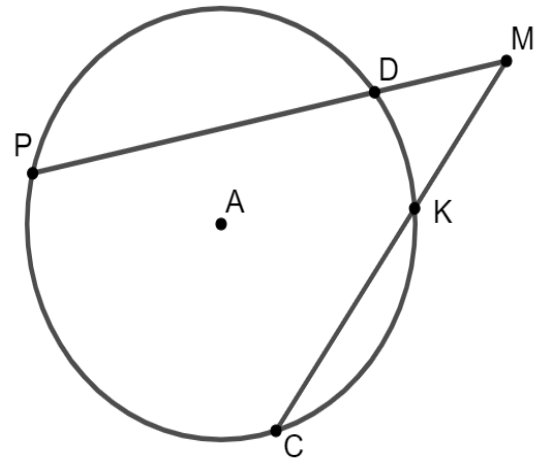
Name _____
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Circle A with secants $\overline{PM}$ and $\overline{CM}$ is shown. $\overline{PM}$ and $\overline{CM}$ intersect at point M outside the circle so that $PD = 17$, $DM = 3$, $CK = x$, and $KM = 5$.

1. What is the length of secant $\overline{PM}$?

2. Find the value of x .

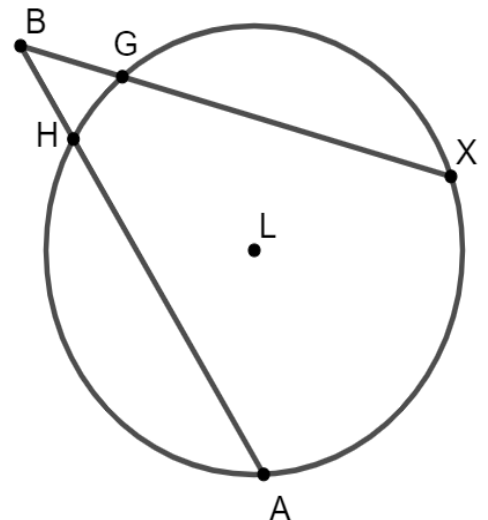
3. What is the length of secant $\overline{CM}$?



Circle L is shown, where points G , X , A , and H lie on circle L . Secants $\overline{BX}$ and $\overline{BA}$ intersect at point B outside the circle such that $BG = 8$, $GX = (7x - 4)$, $BH = 6$, and $HA = 18$.

4. What is the value of x ?

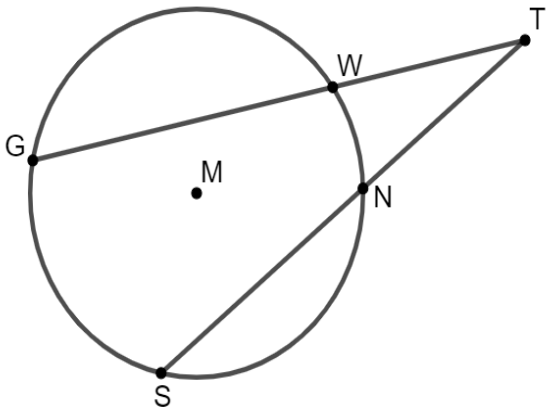
5. Calculate the length of secant $\overline{BX}$.



Circle M with secants $\overline{GT}$ and $\overline{ST}$ is shown. Secants $\overline{GT}$ and $\overline{ST}$ intersect outside the circle at point T so that $GW = 7x$, $WT = 2x$, $SN = 6$, and $NT = 6$.

6. Find the value of x .

7. What is the length of secant $\overline{GT}$?



Bailey and Luciano were solving for x in circle I where points J , M , R , and K are on the circle and secants $\overline{PM}$ and $\overline{PR}$ intersect at point P outside the circle. They know $PJ = 2$, $JM = (12x - 5)$, $PK = 3$, and $KR = 4x$.

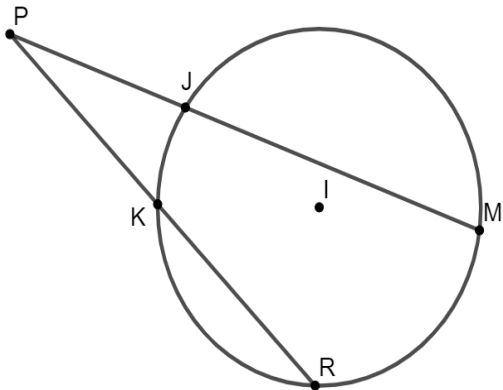
Each of their equations are set up to solve for x .

Bailey's Work	Luciano's Work
$3(4x + 3) = 2(12x - 3)$	$4x(4x + 3) = 2(12x - 3)$

8. Who wrote the correct equation needed to solve for x ?

9. Use this equation to find the value of x .

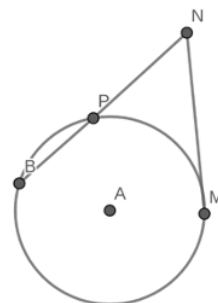
10. What is the length of secant $\overline{PR}$?



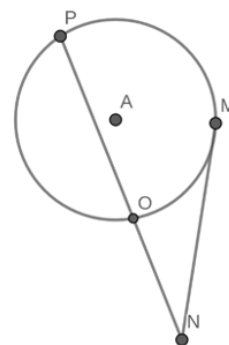
Geometry
Unit 12
Lesson 3 Practice

Name _____
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1. Circle A is shown with points M, B and P on the circle. Secant $\overline{BN}$ and tangent $\overline{MN}$ intersect at point N outside the circle. $PB = 5$ and $PN = 4$. What is the length of $\overline{MN}$?

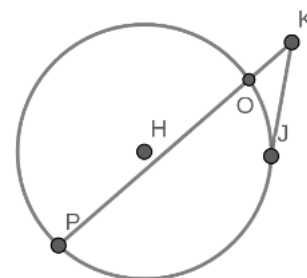


2. Circle A is shown with points M, O and P on the circle. Secant $\overline{PN}$ and tangent $\overline{MN}$ intersect at point N outside the circle. $PO = (3x + 3)$, $ON = 10$ and $MN = 20$. Find the value of x .



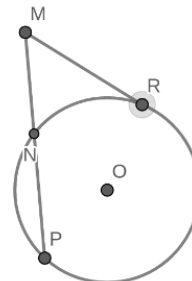
3. What is the length of $\overline{PN}$?

4. Circle H is shown with points J, O and P on the circle. Secant $\overline{KP}$ and tangent $\overline{KJ}$ intersect at point K outside the circle. $PO = 112$, $OK = 14$ and $KJ = x$. Find the value of x .



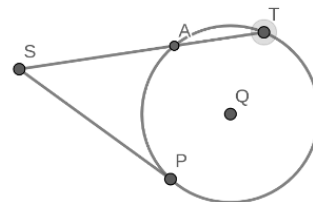
5. Using the same image as question 4, circle H is shown with points J, O and P on the circle. Secant $\overline{KP}$ and tangent $\overline{KJ}$ intersect at point K outside the circle. $PO = 16 + x$, $OK = 2$ and $KJ = 8$. Find the length of $\overline{KP}$.

6. Circle O is shown with points N, R and P on the circle. Secant $\overline{MP}$ and tangent $\overline{MR}$ intersect at point M outside the circle. $PN = 20$, $MN = 16$ and $MR = 4x$. What is the value of x ?



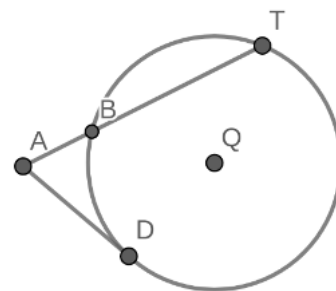
7. Using the same image as question 6, Circle O is shown with points N, R and P on the circle. Secant $\overline{MP}$ and tangent $\overline{MR}$ intersect at point M outside the circle. $PN = x + 10$, $MN = 15$ and $MR = 30$. What is the value of x ?

8. Circle Q is shown with points A, T and P on the circle. Secant $\overline{ST}$ and tangent $\overline{SP}$ intersect at point S outside the circle. $SP = 4x$, $AT = 18$ and $AS = 32$. What is the value of x ?



9. Daniella and Taylor are working on finding the length of $\overline{AD}$. Circle Q is shown with points B, T and D on the circle. Secant $\overline{AT}$ and tangent $\overline{AD}$ intersect at point A outside the circle. $BT = 12$, $AB = 10$ and $AD = 2x$. Their partial work solving for x is shown below. Find the value of x for each of their work. Round to the nearest hundredth. Whose work do you agree with?

Daniella's work	Taylor's work
$(2x)^2 = 10(10 + 12)$ $4x^2 = 10(22)$ $x^2 = 55$	$(2x)^2 = 12(10 + 12)$ $4x^2 = 12(22)$ $x^2 = 132$

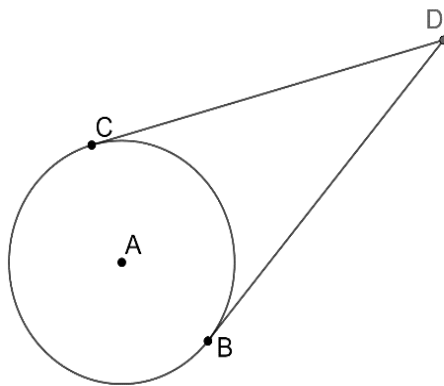


10. Using the image in Question 9, what is the length of $\overline{AD}$?

Geometry
Unit 12
Lesson 4 Practice

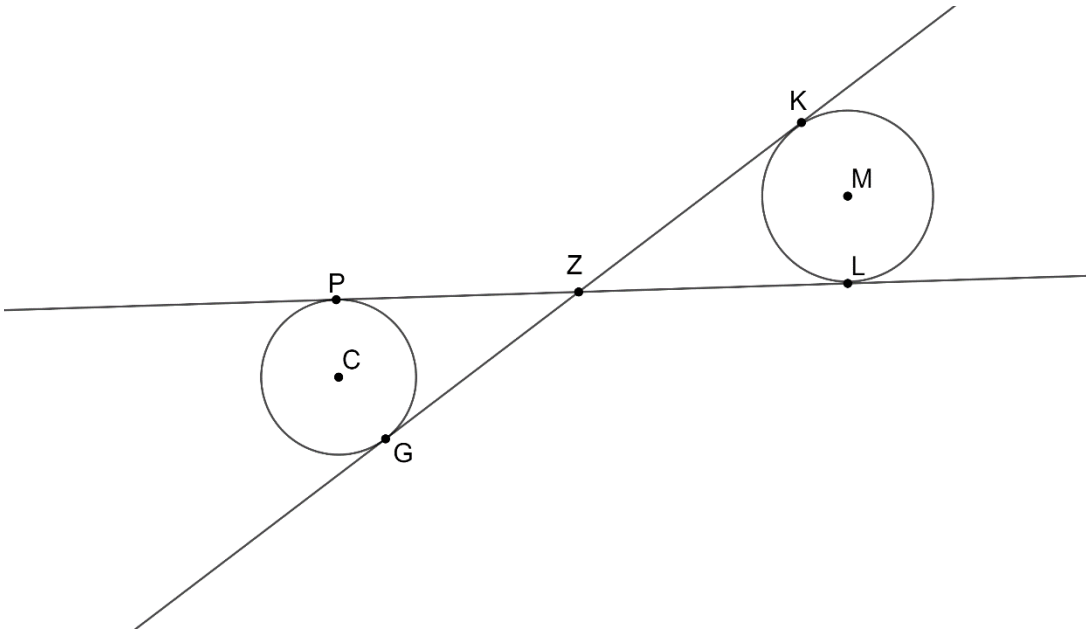
Name _____
Date _____
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For questions 1-3, Circle A is shown. $\overline{CD}$ and $\overline{DB}$ are tangent to the circle.
 $CD = 8x - 20$ and $DB = 13x - 45$.



1. What is the value of x ?
2. What is the length of $\overline{CD}$?
3. What is the length of $\overline{DB}$?

For questions 4-10, Circles C and M are shown. Lines $\overline{PL}$ and $\overline{GK}$ intersect at point Z . Line $\overline{PL}$ is tangent to circle C at point P and circle M at point L . Line $\overline{GK}$ is tangent to circle C at point G and circle M at point K . $PZ = 10y - 3$, $ZL = 4x + 10$, $GZ = 7y + 21$, $ZK = 6x - 16$.

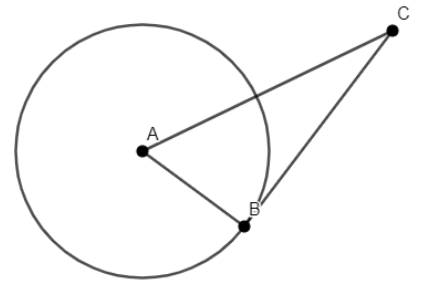


4. Write an equation you can use to solve for x .
5. Solve the equation you wrote in question 4 for x .
6. Write an equation you can use to solve for y .
7. Solve the equation you wrote in question 6 for y .
8. What is the length of $\overline{GK}$?
9. What is the length of $\overline{GZ}$?
10. What is the length of $\overline{ZL}$?

Geometry
Unit 12
Lesson 5 Practice

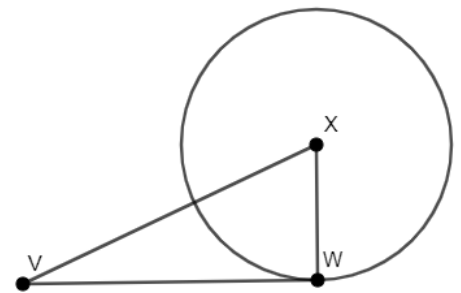
Name _____
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1. Circle A is shown with point C outside the circle, where point B is on the circle, $AC = 26$, $CB = 24$, and $BA = 10$. Verify if $\overline{BC}$ is tangent to circle A using Pythagorean Theorem.



Circle X on the right has $\overline{VW}$ as a tangent segment where W is the point of tangency. $VX = 17$ and $VW = 15$.

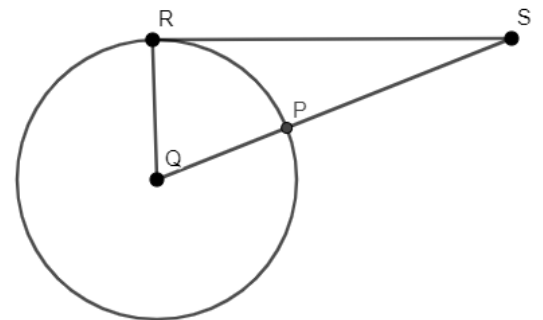
2. What is the length of $\overline{XW}$?



3. What is the area of $\triangle XWV$?

In Circle Q , $\overline{RS}$ is tangent to the circle at point R , $QR = 6$, and $PS = 4$.

4. What is the length of $\overline{RS}$?



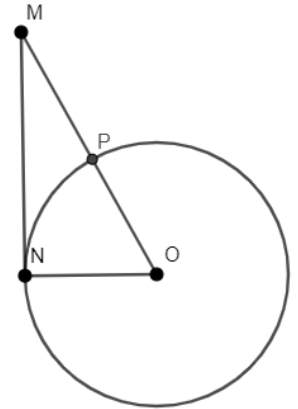
5. What is the area of $\triangle QRS$?

In Circle O , $\overline{MN}$ is a tangent segment whose point of tangency is point N . $MN = 8$ and $MP = 2$.

6. What is the length of $\overline{ON}$?

7. What is the length of $\overline{MO}$?

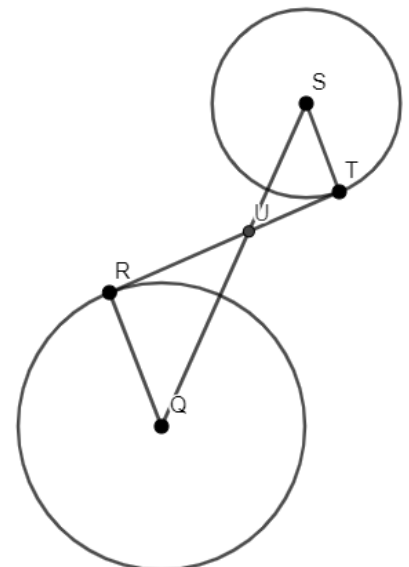
8. What is the perimeter of $\triangle MNO$?



In the figure on the right, $\overline{RT}$ is a common tangent segment of both Circle Q and Circle S . $QR = 5$, $RU = 12$, and $TU = 3$.

9. What is the length of $\overline{ST}$?

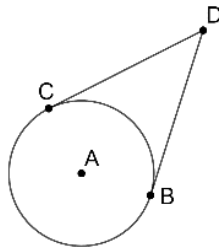
10. What is the length of $\overline{QS}$?



Geometry
Unit 12
Lesson 6 Practice

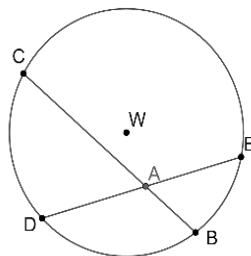
Name _____
Date _____
Period _____

For questions 1 through 3, Circle A is shown. $\overline{CD}$ and $\overline{DB}$ are tangent to the circle.
 $CD = 10x + 8$ and $DB = 15x - 12$.



1. What is the value of x ?
2. What is the length of $\overline{CD}$?
3. What is the length of $\overline{DB}$?

For questions 4 through 7, Circle W is shown. Chords $\overline{CB}$ and $\overline{DE}$ intersect at point A .
 $CA = 4x - 5$, $AB = 4$, $DA = 6$, and $AE = 2x + 2$.



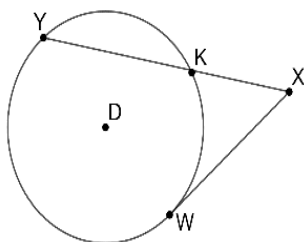
4. Write an equation you can use to solve for x .

5. Solve the equation you wrote in question 4 for x .

6. What is the length of $\overline{CB}$?

7. What is the length of $\overline{DE}$?

For questions 8 through 10, Circle D is shown. $\overline{XY}$ is a secant to the circle and intersects tangent $\overline{XW}$ at point X . $XW = 10$, $KX = 4$, and $YX = 5x + 10$.



8. Write an equation that can be used to find the value of x .

9. Solve the equation you wrote in question 8 to find the value of x .

10. What is the length of $\overline{YK}$?

Geometry
Unit 12
Lesson 7 Practice

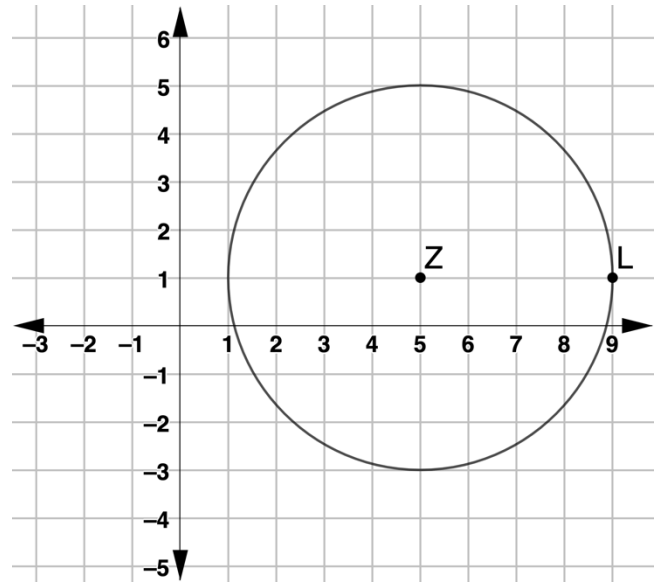
Name _____
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For questions 1 and 2, Circle Z is shown. Point L lies on the circle.

1. Give the key features of the circle.

Center	Diameter

2. Write the equation that represents Circle Z .

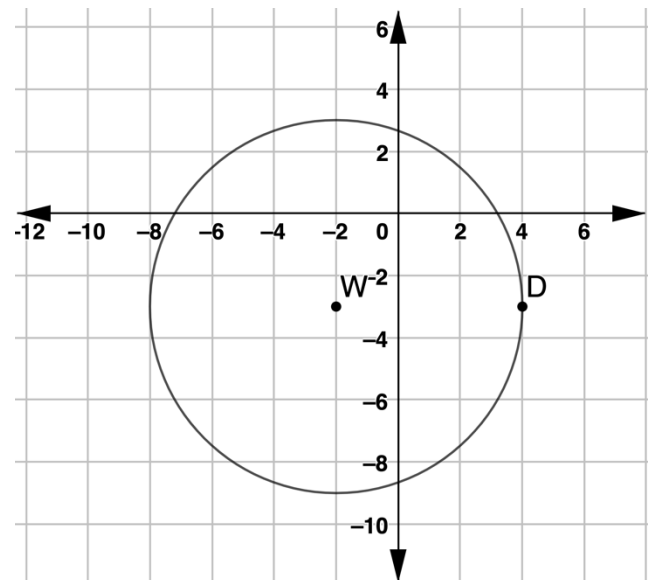


For questions 3 and 4, Circle W is shown. Point D lies on the circle.

3. Give the key features of the circle.

Center	Diameter

4. Write the equation that represents Circle W .

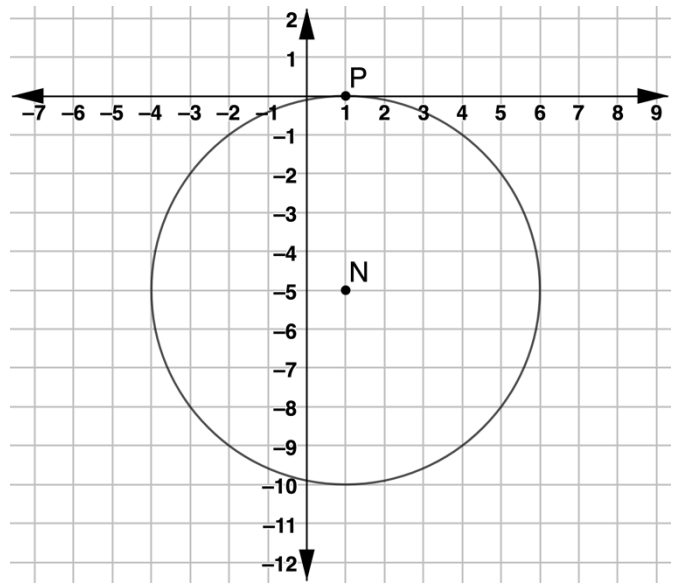


For questions 5 and 6, Circle N is shown. Point P lies on the circle.

5. Give the key features of the circle.

Center	Diameter

6. Write the equation that represents Circle N .

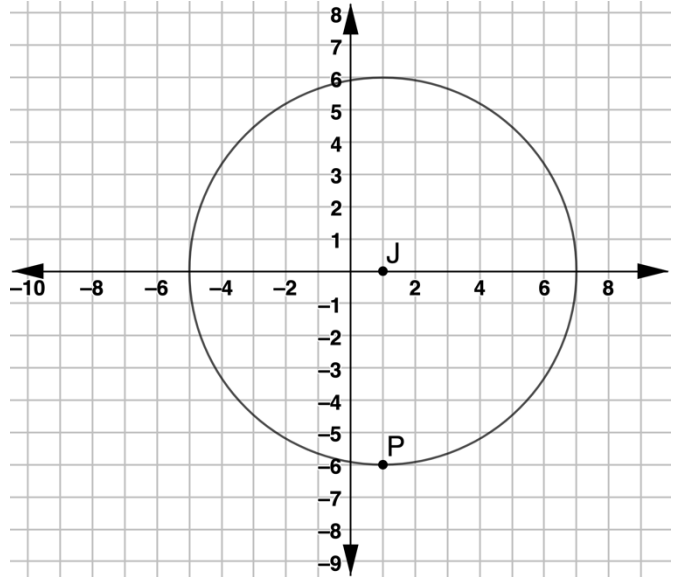


For questions 7 and 8, Circle J is shown. Point P lies on the circle.

7. Give the key features of the circle.

Center	Diameter

8. Write the equation that represents Circle J .



Geometry
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Lesson 8 Practice

Name _____
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For questions 1 and 2, consider the equation of the circle represented by

$$(x - 8)^2 + (y - 10)^2 = 12^2.$$

1. Rewrite the equation in general form.
2. Give the key features of the circle.

Center	Diameter	Radius
Domain	Range	

For questions 3 and 4, consider the equation of the circle represented by

$$(x + 10)^2 + (y - 3)^2 = 1^2.$$

3. Rewrite the equation in general form.
4. Give the key features of the circle.

Center	Diameter	Radius
Domain	Range	

For questions 5 and 6, consider the equation of the circle represented by

$$x^2 + y^2 - 18y = -65.$$

5. Rewrite the equation in center-radius form.
6. Give the key features of the circle.

Center	Diameter	Radius
Domain	Range	

For questions 7 and 8, consider the equation of the circle represented by

$$x^2 + y^2 + 10x = -21.$$

7. Rewrite the equation in center-radius form.
8. Give the key features of the circle.

Center	Diameter	Radius
Domain	Range	

For questions 9 and 10, consider the equation of the circle represented by

$$x^2 + y^2 - 2x + 10y = 23.$$

9. Rewrite the equation in center-radius form.
10. Give the key features of the circle.

Center	Diameter	Radius
Domain	Range	

Geometry
Unit 12
Lesson 9 Practice

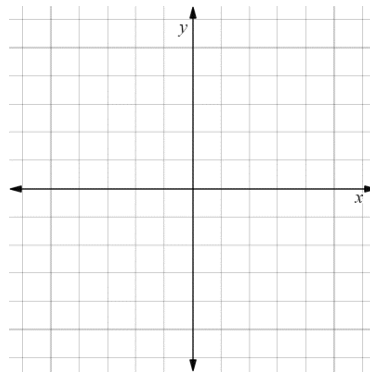
Name _____
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For questions 1 and 2, consider that the equation of circle B is $(x + 4)^2 + (y + 4)^2 = 1$.

1. Give the key features of the circle.

Center	Radius
Domain	Range

2. Graph circle B .

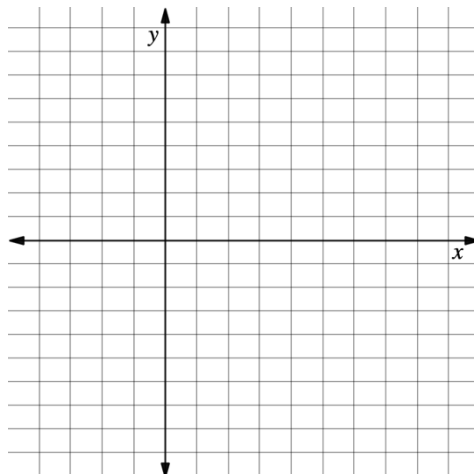


For questions 3 and 4, consider that the equation of circle J is $(x - 5)^2 + (y + 2)^2 = 16$.

3. Give the key features of the circle.

Center	Radius
Domain	Range

4. Graph circle J .



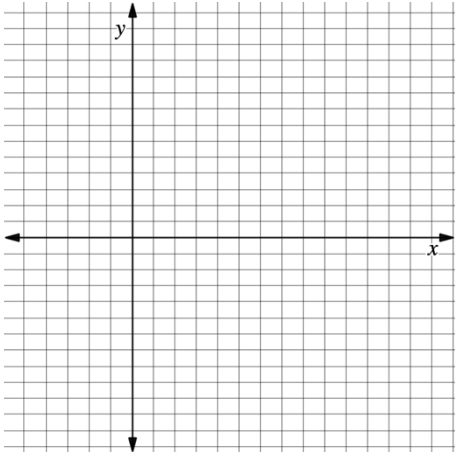
For questions 5 through 7, consider that the equation of circle B is

$$x^2 - 10x + y^2 - 6y = 30.$$

- 5. Rewrite the equation of the circle in center-radius form.
- 6. Give the key features of the circle.

Center	Radius
Domain	Range

- 7. Graph circle B .



For questions 8 through 10, consider that the equation of circle Y is

$$x^2 + 2x + y^2 - 10y + 17 = 0.$$

- 8. Rewrite the equation of the circle in center-radius form.
- 9. Give the key features of the circle.

Center	Radius
Domain	Range

- 10. Graph circle Y .

